

Introduction

Hearing loss affects more than **430 million people worldwide** and is commonly caused by damage to stereocilia, the microscopic hair-like structures located on inner ear hair cells (Corwin 1985). These structures are responsible for **converting sound vibrations into electrical signals** that are interpreted by the brain. Current treatments, including hearing aids and cochlear implants, can improve hearing ability but **cannot repair or regenerate the damaged cells themselves**. Because of this limitation, researchers continue to explore new systems that may help model or support regenerative processes.

One possible model involves the **fungal mycelium of Pleurotus ostreatus**, which forms dense branching networks of hyphae and actin-based structures similar to those found in stereocilia. These filament-like growth patterns may provide a comparison to the growth of inner ear hair cells if grown on hydrogels. Previous studies have shown that hydrogels are capable of maintaining stability and supporting fungal activity, making them an effective material for growing fungal structures.

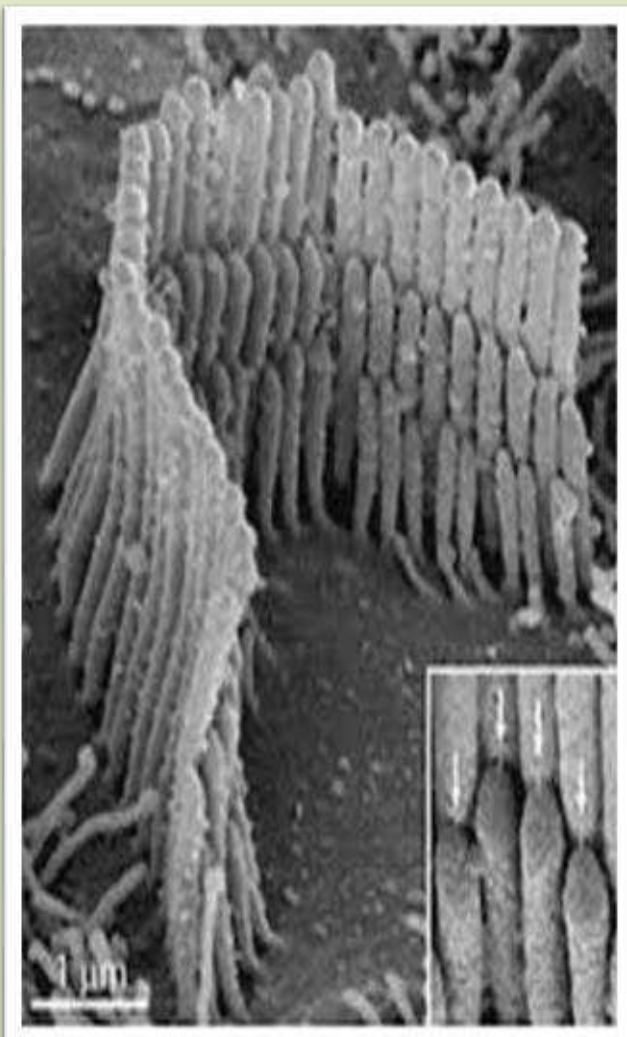


Fig 1. Microscopic Stereocilia

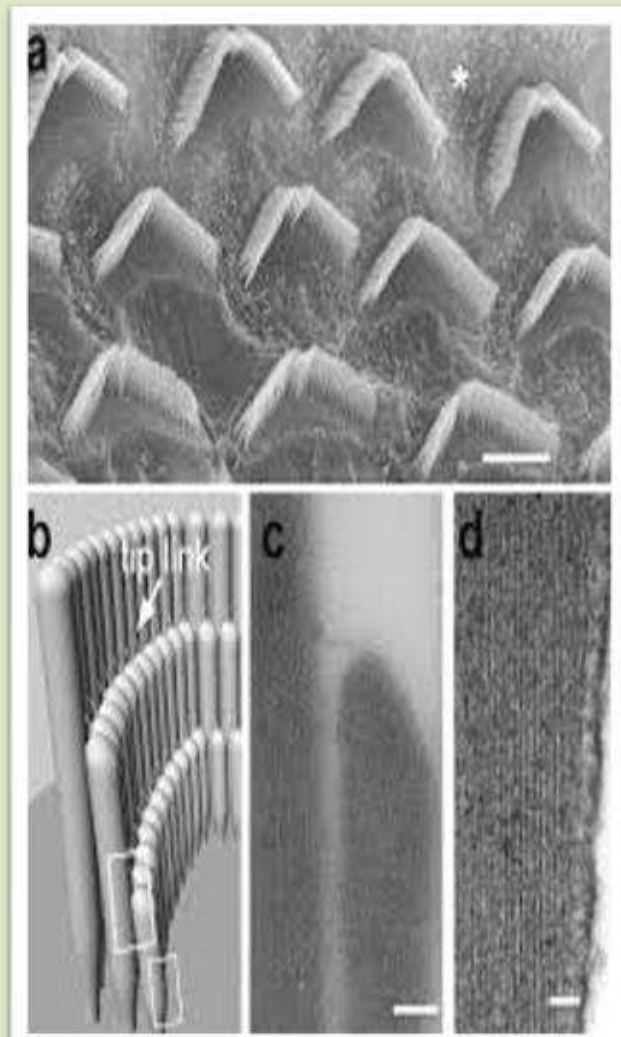


Fig 2. Acting Networks of Stereocilia

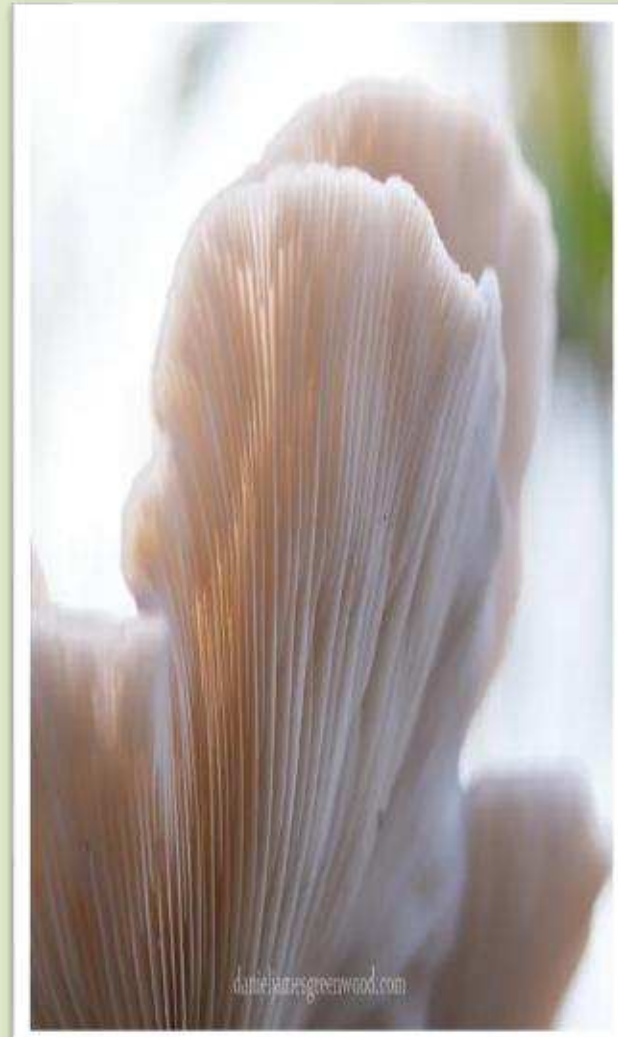


Fig 3. Branching Structures of Fungi

Purpose

Mycelial growth of *Pleurotus ostreatus* on GelMA hydrogels may be used **to model stereocilia regeneration in the inner ear**. By examining how different growth stimulators affect the density and organization of fungal hyphae, the growth patterns are then compared to stereocilia structures associated with hair cell function and hearing loss treatment research.

Evaluating the Effectiveness of Strigolactone and Comfrey on the Growth of *Pleurotus ostreatus* mycelium to Simulate Stereocilia to Reverse Hearing Loss

Methods

1. To create the Hydrogels, gelatin was **set in cold water** for five minutes. This was stirred using a hotplate until fully dissolved, poured into molds, and **refrigerated until set**.
2. **Five controls** were created to test growth effects with **hydrogels containment nutrients** and plant extracts. Negative controls had GelMA with fungi only.
3. The hydrogels **were inoculated with *Pleurotus ostreatus*** by applying 100 uL of spore around the hydrogel of equal distance with each other to ensure maximum growth. This container was placed in a dark room to mimic the stereocilia environment.
4. Using 10× digital zoom device , fungal growth was **counted and measured** to determine which growth medium provided the most growth. Observations of growth and texture were also recorded.

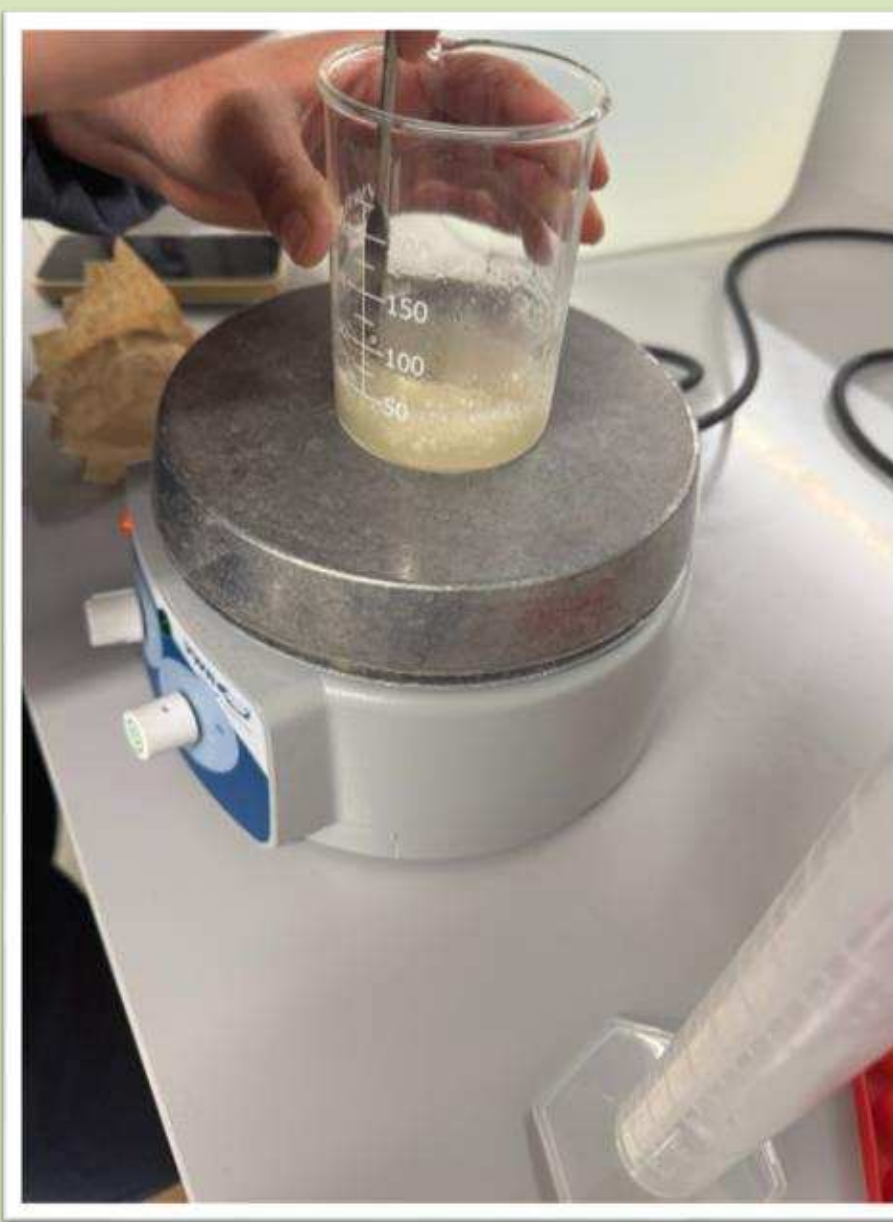


Fig 4. Hydrogel Mixture Before Crystallization of the Hydrogel



Fig 5. Hydrogel Molds after 24 Hours



Fig 6. Container Created to Simulate Inner Ear Environment



Fig 7. Inoculated Hydrogels in Dark Cabinet

Results



Fig 8. Fungal Growth in Control



Fig 9. Fungal Growth in Strigacolone

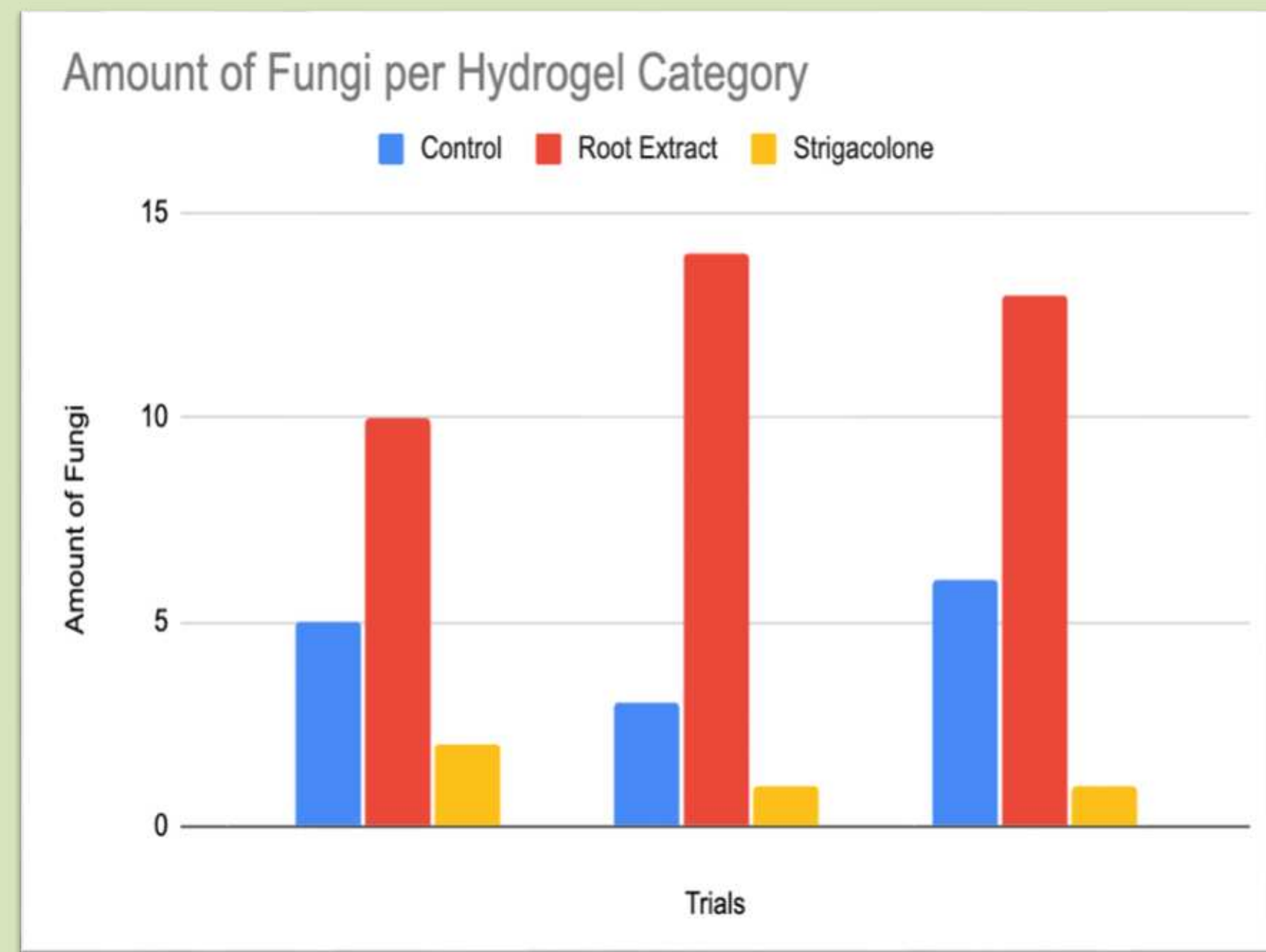


Fig 10. Amount of Fungi per Hydrogel

Conclusion

The growth of *Pleurotus ostreatus* mycelium on hydrogels was examined to determine whether its **structure can resemble stereocilia**. By comparing different conditions, it showed that nutrients and stimulators influence filament length, density, and branching patterns, with the root extract creating the most growth.. The results suggest that fungal mycelium shares **key structural similarities with stereocilia**, particularly in its branching networks and filament organization.

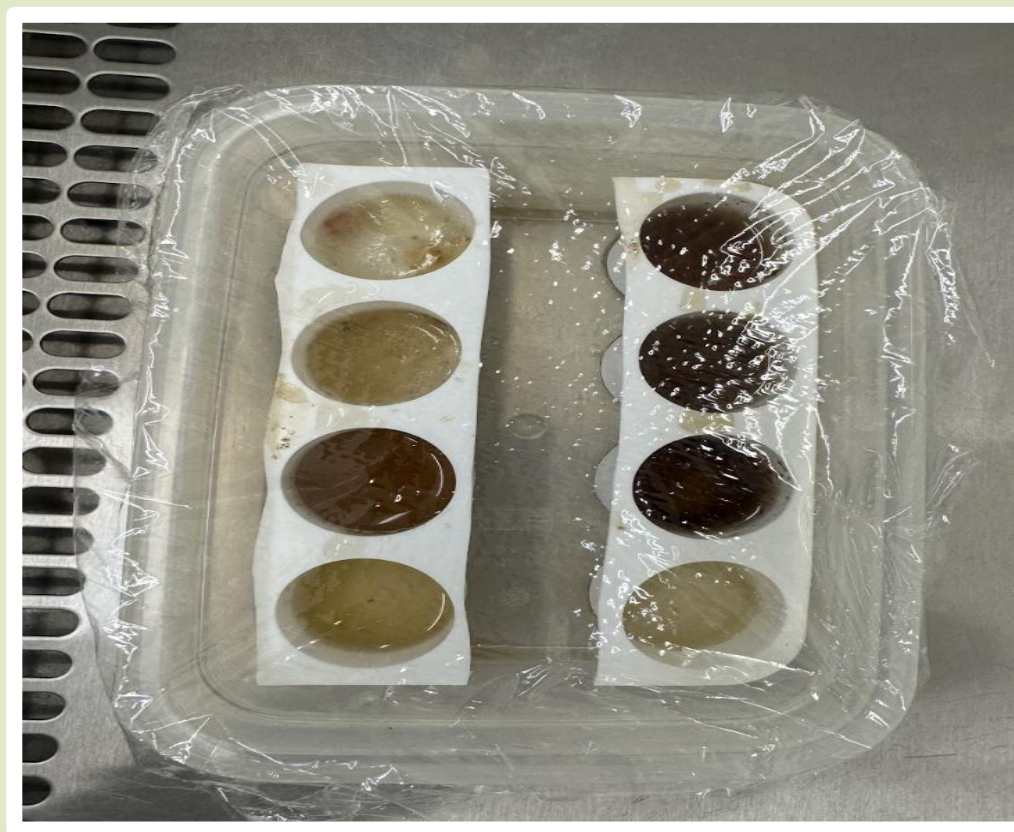


Fig 11. Fungi on Hydrogel after 48 Hours of Inoculation

While it does not replicate stereocilia's function, it can serve as a useful **model** to study how these structures form. *Pleurotus ostreatus* mycelium models stereocilia-like growth, helping improve understanding of filament organization and potentially **informing future research on hearing loss and hair cell regeneration**.

Extensions

By testing **additional growth factors** or stimulators to ensure proper *Pleurotus ostreatus* mycelial structure, stereocilia could be better mimicked. Future experiments could also **explore different hydrogel compositions or stiffness levels** to more closely replicate inner ear tissue. Using advanced imaging techniques, such as **scanning electron microscopy**, could allow for more precise comparisons of filament structure. Additionally, comparing multiple fungal species and incorporating cell cultures may help strengthen the connection between this model and regeneration.

Citations

Bittencourt, A. G., Ikari, L. S., Della Torre, A. A., Bento, R. F., Tsuji, R. K., & Brito Neto, R. V. (2012). Post-lingual deafness: benefits of cochlear implants vs. conventional hearing aids. *Brazilian journal of otorhinolaryngology*, 78(2), 124–127. <https://doi.org/10.1590/S1808-86942012000200019>

Corwin, Jeffrey T. "Perpetual Production of Hair Cells and Maturation Changes in Hair Cell Ultrastructure Accompany Postembryonic Growth in an Amphibian Ear." *Proceedings of the National Academy of Sciences of the United States of America*, vol. 82, no. 11, 1985, pp. 3911–15. JSTOR, <http://www.jstor.org/stable/25781>.

"Stereocilia - an Overview | ScienceDirect Topics." www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/stereocilia.